ECPP integration: guarded engineering checks

27 October 2026 | TASK-20261027-275b12

Observed checkpoint: all 36 tests pass on the third full attempt. No real VM, original program, scientific cohort or storage operation ran. Historical selection and the full commonality objective remain unresolved.

Attempt	Tests	Failures	Errors	Skipped
1	36	1	51	0
2	36	2	0	0
3	36	0	0	0

Additive ECPP integration implementation

This engineering report covers TASK-20261027-275b12, EXP-NISTSEED-6886da, GOAL-NISTSEED-ab8949, BATCH-a41798. The original source-authoring narrative below is retained with its prospective labels. The operator checkpoint appended at the end records actual tests and corrections. No historical explanation, scientific admission, hypothesis disposition or goal completion is established.

The original objective remains an independently reviewed historical procedure explaining the reconstructed omissions and published controls, or a prospectively frozen independently validated predictive property on its declared holdout domain. Integration source, controls and any later ECPP observations do not replace that criterion.

Source review and bindings

Provenance: internal. Verified_by: the source-authoring Executor read the committed integration handoff and INTEGRATION_CONTRACT.md; frozen V9 runner/checker; capsule.py and capsule_check.py; printed_trace.py; the existing A/A producer, schema and tests; qemu_controls.py image/vm code; qemu_init.c; guest_framing.py; certificate_admission.py; source-inputs.json; and research-visuals/PDF skill requirements. The Coordinator's earlier source/literature review and its acquired-source limitations remain those recorded in INTEGRATION_CONTRACT.md. This authoring pass did not independently reacquire the historical archive or thesis and makes no novelty or historical-byte identity claim.

Source-inputs.json is bound independently in both new modules by SHA-256 47341a3e609a595e80c82042029377eec129dc9a29347bf75430e1e1d4339f62. It identifies source commit 4c7298e5212f523642c782483cd28f4bd5000d66 and all 23 inputs. Additional source pins are literal constants in both modules. The full frozen ordered cohort and independent prior mapping are compared, including all 21 UIDs, 11 omitted and 10 published rows. Complete row identity is retained; no representative or candidate universe is reduced.

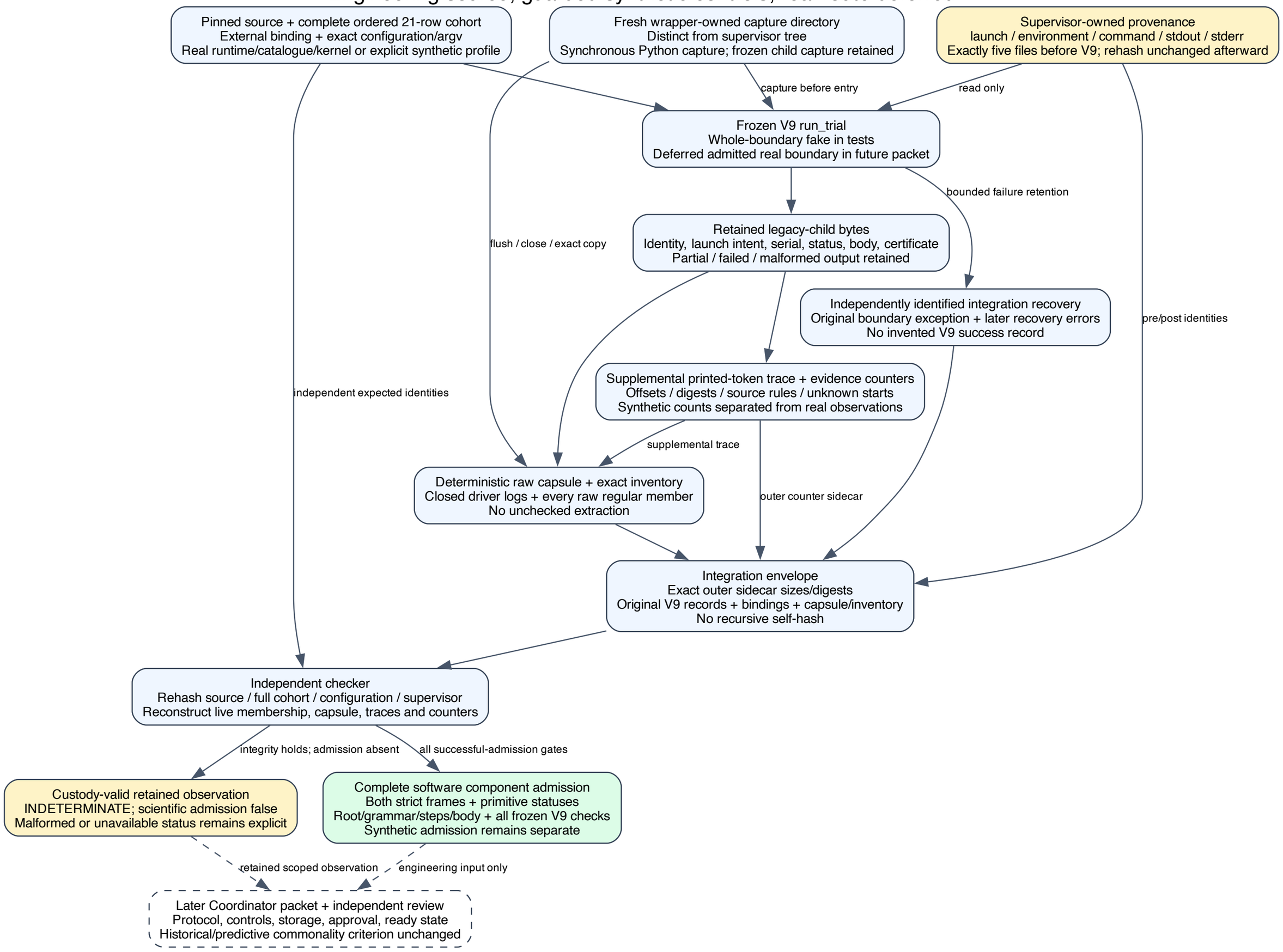
The A/A producer remains the existing source with SHA-256 d481423ee4562c345d575f1dbd8f57a1dd5184e2c599c813a2f91154e790ef18. The new code does not implement a replacement producer. Tests call its build_pair with an injected whole-boundary fake. A separately declared synthetic runtime environment supplies tiny cache/image bytes for that control; these bytes are never real runtime admission.

Implementation design

The runner exposes an injected whole-boundary seam and a deferred real route around the frozen V9 default. Importing the new modules selects no boundary. The real route requires actual source/runtime/catalogue/kernel inputs, exact pure image reconstruction and an external admitted-packet callback receipt bound to the full request. That callback remains the later packet's responsibility; a callback receipt is not a scientific decision issued by this implementation.

The independent checker does not import integrated_runner. It rehashes the pinned sources, rederives the full prior map, verifies external/copied bindings and the exact five supervisor files, enumerates live custody, reuses the independent capsule reader, and additionally reconstructs the deterministic capsule bytes without extraction. Every outer sidecar is bound by size

TASK-20261027-275b12: additive ECPP custody flow
Engineering source; guarded synthetic controls; real route deferred



and digest in the nonrecursive integration envelope. The original V9 envelopes remain byte-preserved.

Capture files start in a distinct fresh declared directory outside the supervisor directory. The runner captures Python stdout/stderr synchronously and closes those handles before copying exact bytes into the trial directory. The real frozen QEMU subprocess retains its existing capture_output mechanism. This is not a general OS descriptor capture facility and cannot safely be shared by concurrent Python threads. The implementation and tests require synchronous operation without background writes.

Finalization closes captures, preserves original V9 records, derives supplemental printed-token traces, records recovery/counters, constructs the raw capsule/inventory, rehashes supervisor provenance and binds the outer sidecars. If finalization fails, the wrapper records an incomplete envelope and preserves captured material. A second recovery error is guarded separately. The boundary wrapper preserves the original boundary exception before a frozen V9 recovery parser can raise another exception.

Successful component admission requires complete independently parsed prover and checker frames, exact ABI and marker counts/order, reconciled primitive status, QEMU exit 33, child exit zero, no watchdog, exact launch/input/runtime identities, one prime label, one root-bound certificate, certificate custody, original V9 trace/static/prior checks and checker root/grammar/step/body checks. A vm_valid literal cannot compensate for malformed checker framing.

Malformed output can remain custody-valid while classified INDETERMINATE, with no scientific admission. Missing status is unavailable. Raw serial digests, lengths and evidence offsets remain descriptive. Unparseable fragments are not relabeled complete bodies or certificates. Supplemental traces preserve printed tokens, repeats, incomplete lines and unknown intervening text; unprinted state remains UNOBSERVED.

Counters separate selected indices, whole-boundary attempts, launch intent, guest initialization, source-backed output signatures, successful/failed completed stages and independent certificate admission. Exact original-program starts remain unknown where the retained events do not discriminate them. The original named_orders_executed literal is labeled legacy_selected_work_field and is never summed as program starts. Synthetic evidence contributes nothing to real execution totals.

Prospective verification and actual execution status

No tests were executed by this source-authoring Executor. Root reported materializing the first runner chunk and compiling syntax only; this report does not convert that operator report into a completed integration test. The original runner delivery digest was 6855875f42cfe21c9f9e711635b80a24d3b06871b241a378a1f6e61417e1ad40. A guarded correction fixes the root path from HERE.parents[5] to HERE.parents[4], binds the separate capture path, preserves boundary exceptions and permits exact retained checker-input bytes to support launch identity during malformed-frame retention.

Pending actual command, from the checkout: PYTHONDONTWRITEBYTECODE=1 python3 experiments/EXP-NISTSEED-6886da/integration/tasks/TASK-20261027-275b12/test_integration.py. Root must retain command, environment/tool versions, exit status, complete stdout/stderr, actual tests_run/failures/errors/skipped and every failed attempt in test-report.txt. The suite reports instrumented mock calls separately and writes its fixture artifacts to distinct task-local _mock_artifacts directories without overwriting or deleting failed attempts. Generated fixture directories must remain available for diagnosing failures; the declared snapshot artifacts remain the nine paths in the handoff.

Pending controls cover complete admission; early failure; launch-only and guest-before-exec failure; partial original-program output; independently bounded recovery exceptions; absent/duplicate/misordered/malformed markers and ABI; truncated certificate; malformed successful-looking checker output; incorrect checker step/root; prime/composite labels; watchdog/nonzero child; UID/order/label/index and source/runtime/config/argv/input identities; copied binding and supervisor logs; changed/missing/extra/symlinked raw members; capsule/inventory/sidecar and supplemental-trace tampering; invented counts; reused/aliased destinations; separate A/A validity before comparison; and an actual local explicit-fake quiet CLI subprocess inspected after process exit. These are planned controls until the retained receipt records their outcomes.

Actual real execution totals in this authoring session: VM 0; legacy program 0; scientific cohort run 0; storage operation 0. Actual tests in this authoring session: 0. Mock and local-CLI process totals are presently unmeasured and must come from Root's actual test receipt. No performance benchmark or mathematical measurement is requested or reported.

Ownership and deviations

The first required Conductor scope check returned block_conflict because T-5464 owned the parent integration subtree. The Executor stopped before editing. Root subsequently retained materialization in its accepted task context and requested read-only source delivery. One read-only cat source retrieval preceded Root's clarification that read-only inspection commands were allowed; Root was notified. The child has made no filesystem writes, project-code imports, test invocations, commits, storage calls or scientific-state changes.

The current delivery is source proposed for Root to materialize and test. The independent engineering validator TASK-20261027-344233 has not been performed or claimed here. The exact archival owner remains TASK-20261027-2ce9dc. Every actual test failure, correction and resulting source digest must be retained before that snapshot.

Visual record and remaining gates

custody.dot is an engineering artifact-flow diagram. Its SVG rendering must be produced from that exact source and included in report.pdf. The runner contains a pure render_report entry point that consumes this Markdown report, the actual test-report.txt and custody.svg. Root must run the PDF operation marker before first PDF authoring, render every PDF page, visually inspect all pages and record that inspection. Neither SVG nor PDF exists merely because its source is proposed here.

Canonical graph review is pending Root's actual read of the affected study/frontier graph sources. This engineering design adds no mathematical, transfer or attack-cost edge; no canonical graph change is proposed. The final report must name the graph paths actually checked, with their digests and why unchanged, rather than asserting an unperformed check.

The later full packet still requires actual runtime availability, durable artifact-store authorization, the complete experiment.protocol/v1 and controls, variable/sampling/analysis/checkpoint policies, experiment.export/v1, complete supervisor/trial archive coverage, independent replay/review, preapproval true, committed Coordinator approval and ready-state confirmation. Current missing runtime files must be reported as exact admission limits. A passing synthetic suite cannot establish real-runtime admission or the historical/predictive commonality criterion.

Operator checkpoint, 27 October 2026

The complete source was materialized from public Executor deliveries, with original and revised hashes retained. Root-path and capture-binding fixes, original boundary-exception retention and retained checker-input handling were applied before tests. The schema timestamp field was corrected to the integer encoding in all 21 pinned rows. Two test assertions were then corrected to require earlier intended refusal gates; no checker protection was relaxed.

The guarded import/root control passed. Three complete integration suites were attempted. Attempt 1: 36 tests, 1 failure, 51 errors, 0 skips. Attempt 2: 36 tests, 2 failures, 0 errors, 0 skips. Attempt 3: 36 tests, 0 failures, 0 errors, 0 skips, exit 0. The successful run reports 75 injected boundary calls, one explicit fake CLI process and zero forbidden calls. All real VM, legacy program, scientific cohort and storage execution totals remain zero. Exact source hashes, commands, host/tool identities, timestamps and complete stdout/stderr for all attempts are losslessly retained in test-report.txt. Its original 161035 bytes are gzip+base64 encoded with SHA256 4322566bfc64ca745d43921d37922991f93220a07d92491069ae96e81abf19f1 and round-trip byte equality checked. This keeps the declared report artifact below storage thresholds without discarding raw errors. Task-local fixture directories remain preserved and are not claimed as archived by a source-only nine-file snapshot.

The actual command for each full suite was PYTHONDONTWRITEBYTECODE=1 /Users/adamburan/.pyenv/versions/3.12.8/bin/python3 experiments/EXP-NISTSEED-6886da/integration/tasks/TASK-20261027-275b12/test_integration.py. External watchdog: 180 seconds. The native Python is 3.12.8 with jsonschema 4.26.0 on macOS-26.6-arm64-arm-64bit. No comparative timing or performance claim is made.

Canonical sources actually read and checked, with SHA256:

- research/nist_seed_ecpp_compat_20261024/runtime.dot: 26f8781ba81dfa3d38e0029a9879974e3df057cc6a1b86e77cdf6204190844ae - research/nist_seed_ecpp_compat_20261024/admission.dot: 815692bf49dbafffdd20a70879c90f0950334acb713a53c3a22731380972f13f - knowledge/frontiers/ecdlp/README.md: 4e69075f2802dfae42baf748b918d85f6b1d566dd8a57ad347a0a5c49cfdeffe

No mathematical or attack-cost edge changes. The runtime graph still correctly marks named replay unexecuted. The admission graph keeps integration unfinished until source archival and independent engineering review; this self-test

checkpoint does not resolve that gate. The new custody diagram depicts software artifact flow only.

Remaining: exact nine-file source snapshot and dispatcher-verifiable archive; independent engineering validation; then a lawful full-cohort scientific packet and/or the separately frozen fixed-x test admission. Raw storage access and historical caller evidence remain separate unresolved requirements. Passing this suite does not classify any named order or explain any omission.